

Certificate of Calibration

Report Number: 935212

Sensor Model: CX-1080-CU-HT-20L	Serial Number: X115143
Sensor Type: Cernox Resistor	Calibration Date: October 12, 2016
Sensor Excitation: see <i>Test Data</i> page of report	Calibration Due:
Temperature Range: 20.0 K to 325 K	

Traceability and Calibration Method

This temperature sensor has been calibrated to the International Temperature Scale of 1990 (ITS-90) or the Provisional Low Temperature Scale (PLTS-2000) as appropriate. The calibrations are traceable to the National Institute of Standards and Technology (NIST, United States), the National Physical Laboratory (NPL, United Kingdom), the Physikalisch-Technische Bundesanstalt (PTB, Germany), or natural physical constants.

Lake Shore Cryotronics maintains ITS-90 and PLTS-2000 on standard platinum (PRT), rhodium-iron (RIRT), and germanium (GRT) resistance thermometers that have been calibrated directly by an internationally recognized national metrology institute (NIST, NPL, PTB) for $T < 330$ K or an ISO 17025 accredited metrology laboratory for $330 \text{ K} < T < 800$ K. A nuclear orientation thermometer is also used for temperatures less than 50 mK. These standards are routinely intercompared to verify consistency and accuracy of the temperature scale.

The sensor calibrations are performed by comparison to laboratory standard resistance thermometers and tested in accordance with Lake Shore Cryotronics, Inc. Quality Assurance Manual (QP-4220). The quality system of Lake Shore Cryotronics is registered to ISO 9001.

Procedures used: 021-97-02, 099-00-00, 121-96-02, 029-95-02

Notes

The calibration results in this report apply only to the specific sensor specified above.

This report shall not be reproduced, except in full, without written approval from Lake Shore Cryotronics, Inc.

Unless stated otherwise, the uncertainties in this report are based on an approximate 95% confidence level with a coverage factor $k=2$.

Reported by: Matt Vance
Calibration
Engineer/Technician

Approved by: Romerero Prince
Metrology



DATA PLOT

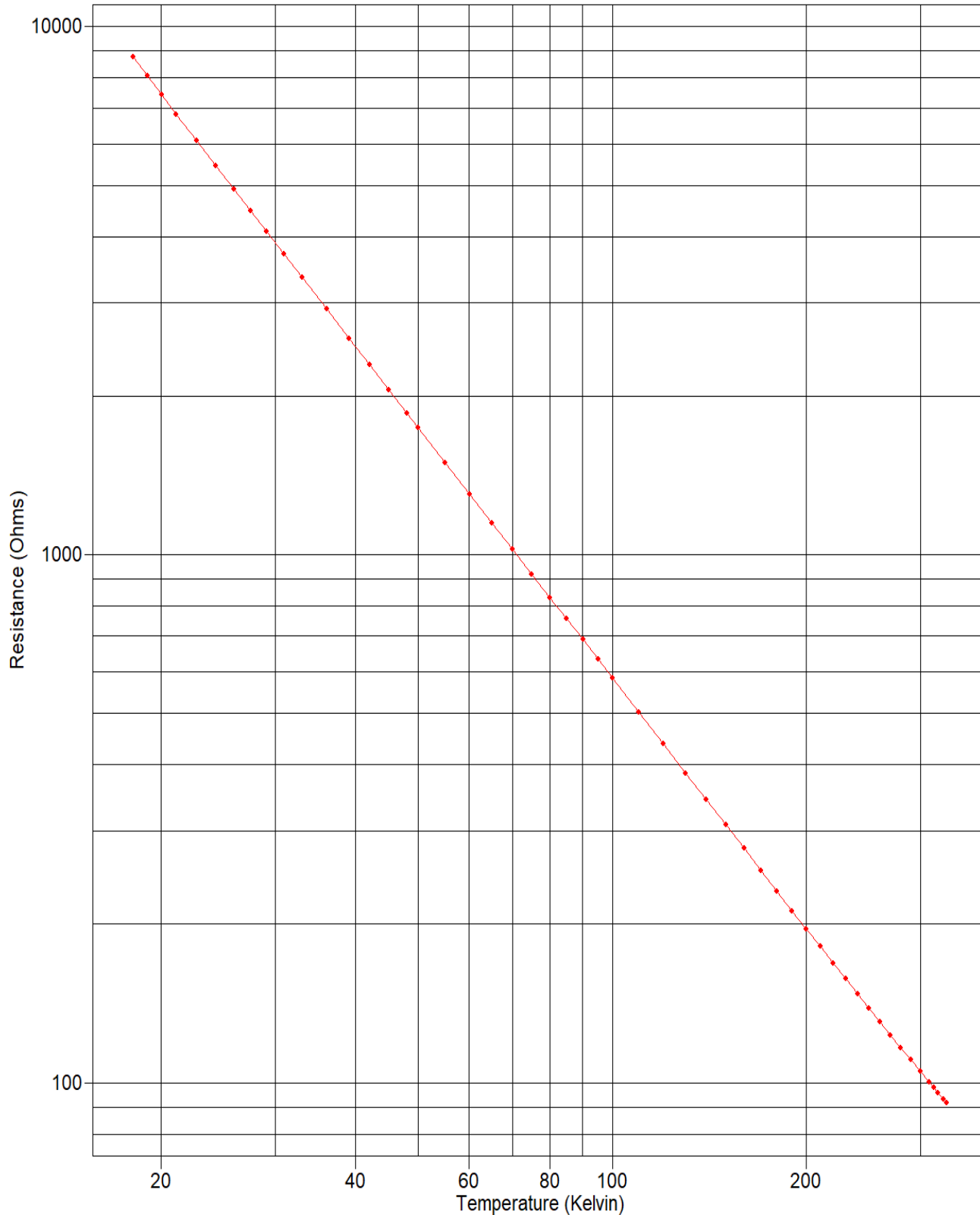
Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K



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TEST DATA

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Index	Temp. (K)	Resistance (Ω)	Excitation	Index	Temp. (K)	Resistance (Ω)	Excitation
1	18.0798	8774.34	2mV $\pm$ 25%	31	140.018	342.887	2mV $\pm$ 25%
2	19.0553	8055.77	2mV $\pm$ 25%	32	150.018	307.380	2mV $\pm$ 25%
3	20.0320	7430.41	2mV $\pm$ 25%	33	160.026	277.506	2mV $\pm$ 25%
4	21.1092	6829.23	2mV $\pm$ 25%	34	170.029	252.133	2mV $\pm$ 25%
5	22.7051	6077.65	2mV $\pm$ 25%	35	180.049	230.378	2mV $\pm$ 25%
6	24.2921	5456.38	2mV $\pm$ 25%	36	190.042	211.588	2mV $\pm$ 25%
7	25.8896	4931.15	2mV $\pm$ 25%	37	200.035	195.236	2mV $\pm$ 25%
8	27.5041	4480.30	2mV $\pm$ 25%	38	210.034	180.913	2mV $\pm$ 25%
9	29.1286	4090.85	2mV $\pm$ 25%	39	220.040	168.274	2mV $\pm$ 25%
10	30.9517	3716.22	2mV $\pm$ 25%	40	230.039	157.091	2mV $\pm$ 25%
11	33.0703	3346.88	2mV $\pm$ 25%	41	240.029	147.138	2mV $\pm$ 25%
12	36.0771	2917.18	2mV $\pm$ 25%	42	250.036	138.217	2mV $\pm$ 25%
13	39.0828	2571.12	2mV $\pm$ 25%	43	260.034	130.217	2mV $\pm$ 25%
14	42.0770	2288.68	2mV $\pm$ 25%	44	270.040	122.992	2mV $\pm$ 25%
15	45.0779	2053.00	2mV $\pm$ 25%	45	280.032	116.461	2mV $\pm$ 25%
16	48.0863	1854.28	2mV $\pm$ 25%	46	290.045	110.516	2mV $\pm$ 25%
17	50.0793	1739.29	2mV $\pm$ 25%	47	300.043	105.111	2mV $\pm$ 25%
18	55.0751	1497.03	2mV $\pm$ 25%	48	310.053	100.173	2mV $\pm$ 25%
19	60.0696	1305.55	2mV $\pm$ 25%	49	315.060	97.8633	2mV $\pm$ 25%
20	65.0613	1151.25	2mV $\pm$ 25%	50	320.056	95.6487	2mV $\pm$ 25%
21	70.0618	1024.33	2mV $\pm$ 25%	51	326.051	93.1209	2mV $\pm$ 25%
22	75.0549	918.925	2mV $\pm$ 25%	52	330.054	91.4975	2mV $\pm$ 25%
23	80.0488	830.162	2mV $\pm$ 25%				
24	85.0472	754.463	2mV $\pm$ 25%				
25	90.0422	689.490	2mV $\pm$ 25%				
26	95.0379	633.123	2mV $\pm$ 25%				
27	100.038	583.854	2mV $\pm$ 25%				
28	110.043	502.141	2mV $\pm$ 25%				
29	120.035	437.588	2mV $\pm$ 25%				
30	130.025	385.552	2mV $\pm$ 25%				

UNCERTAINTY ANALYSIS

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Calibration Data Uncertainty

The uncertainties of the measured calibration data for Lake Shore's sensors are summarized in the table below. The values given are the combined uncertainty of the temperature measurement and the resistance or voltage measurement expressed as an equivalent temperature uncertainty in millikelvin (mK). Note that the values are the calibration uncertainty only and do not include the stability of the temperature sensor. The uncertainty analysis has followed the guidelines for determining measurement uncertainty as outlined in the ISO Guide to the Expression of Uncertainty in Measurement, NIST Technical Note 1297, and ANSI/NCSS Z540-2-1997. Since the uncertainty varies with temperature due to the variation of the sensor sensitivity and excitation, the table gives typical values at several different temperatures throughout the range of the calibration. The uncertainty is based on an approximate 95% confidence level with a coverage factor $k = 2$.

T (K)	Uncertainty ($\pm$ mK)												
	GR	Cernox (CX)					RX			Platinum		RF-800	Diode
		1010	1030	1050	1070	1080	102A	103A	202A	100 Ω	25 Ω	27 Ω	
1.4	4	4	4	4			4	4	4			5	7
4.2	4	4	4	4	4		4	6	5			5	5
10	4	5	5	4	4		10	15	12			7	6
20	8	10	9	8	8	8	35	35	28	9	10	13	9
30	9	13	11	9	9	9	76	61	46	9	9	14	31
50	11	18	14	12	12	11				10	10	13	37
100	20	29	22	17	16	14				11	12	12	32
300		78	60	46	45	36				24	24	25	35
400		124	94	74	72	60				45	45	45	49
500										51	51		54

Polynomial Fit Uncertainty

When a sensor is used to measure temperature, a polynomial fit to the measured calibration data is often used to convert the sensor resistance (R) or voltage (V) to a temperature (T). How well the polynomial represents the sensor calibration data is another source of uncertainty when using the sensor. In the polynomials provided with this set of calibration data, the standard deviation of the fit can be used as an estimate of this additional temperature uncertainty. The standard deviation of fit is determined from the following equation:

$$\sigma_{fit}^2 = \frac{\sum_{i=1}^N (T_i - T_{i,calc})^2}{N - n} = \frac{N}{N - n} (\Delta T_{RMS})^2$$

where

σ_{fit} = standard deviation of the fit

T_i = measured temperature for point i

$T_{i,calc}$ = the temperature calculated from the polynomial equation for point i

N = number of data points in fit range

n = number of fit coefficients

ΔT_{RMS} = root mean square deviation of fit

A value of ΔT_{RMS} is given for each range of fit.

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POLYNOMIAL EQUATION

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

20.0 K to 95.0 K
7450 ohms to 633.1 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 2.70082609715

ZU = 3.94321461445

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	54.094213	2.1828E-04	247815.44
1	-44.544197	3.5244E-04	-126388.98
2	9.806251	3.2820E-04	29879.28
3	-1.426749	3.0865E-04	-4622.49
4	0.160925	2.9044E-04	554.08
5	-0.010882	2.8198E-04	-38.59
6	0.000029	2.6763E-04	0.11

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i \cdot \text{COS}(i \cdot \text{ARCCOS}(k))$, where $0 \leq i \leq 6$
and the A_i 's are the coefficients in the table above.

POLYNOMIAL EQUATION

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Temp. (K) vs. Log(Resistance)

	R Meas. (W)	T Meas. (K)	T Eq. (K)	T diff. (mK)
1	8774.343	18.07980	18.07959	0.20
2	8055.766	19.05531	19.05522	0.09
3	7430.408	20.03195	20.03189	0.06
4	6829.228	21.10921	21.11028	-1.07
5	6077.647	22.70510	22.70454	0.56
6	5456.380	24.29215	24.29284	-0.69
7	4931.146	25.88958	25.88961	-0.03
8	4480.303	27.50413	27.50261	1.52
9	4090.851	29.12857	29.12793	0.64
10	3716.219	30.95167	30.95132	0.35
11	3346.876	33.07028	33.07105	-0.77
12	2917.177	36.07706	36.07819	-1.13
13	2571.123	39.08285	39.08407	-1.22
14	2288.678	42.07695	42.07658	0.38
15	2052.996	45.07791	45.07846	-0.54
16	1854.277	48.08630	48.08504	1.26
17	1739.295	50.07932	50.07758	1.74
18	1497.031	55.07512	55.07602	-0.90
19	1305.553	60.06964	60.07075	-1.10
20	1151.245	65.06126	65.05955	1.71
21	1024.328	70.06181	70.06205	-0.25
22	918.9246	75.05492	75.05602	-1.10
23	830.1617	80.04879	80.04833	0.46
24	754.4627	85.04716	85.04864	-1.48
25	689.4901	90.04220	90.04141	0.79
26	633.1230	95.03790	95.03760	0.30
27	583.8540	100.03792	100.03726	0.66
28	502.1415	110.04280	110.04325	-0.45

Order of Fit = 6 RMS error of fit = 0.91 mK

Largest absolute error = 1.74 mK at data point no. 17

POLYNOMIAL EQUATION

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

95.0 K to 325 K
633.1 ohms to 93.55 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 1.96140912834

ZU = 2.87763775617

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	186.078579	8.4058E-04	221368.42
1	-119.325636	1.2883E-03	-92623.79
2	20.932054	1.2180E-03	17185.79
3	-3.079202	1.2190E-03	-2526.11
4	0.524491	1.1759E-03	446.02
5	-0.095225	1.1212E-03	-84.93
6	0.015863	1.1050E-03	14.36
7	-0.004953	1.1207E-03	-4.42
8	0.002880	1.0998E-03	2.62

$Z = \text{Log}(\text{Resistance})$

$k = ((Z - Z_L) - (Z_U - Z)) / (Z_U - Z_L)$

Temp. (K) = $\sum A_i \cdot \text{COS}(i \cdot \text{ARCCOS}(k))$, where $0 \leq i \leq 8$
and the A_i 's are the coefficients in the table above.

POLYNOMIAL EQUATION

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Temp. (K) vs. Log(Resistance)

	R Meas. (W)	T Meas. (K)	T Eq. (K)	T diff. (mK)
24	754.4627	85.04864	85.04885	-0.22
25	689.4901	90.04141	90.04071	0.70
26	633.1230	95.03760	95.03814	-0.54
27	583.8540	100.03792	100.03797	-0.05
28	502.1415	110.04280	110.04270	0.10
29	437.5885	120.03482	120.03554	-0.72
30	385.5525	130.02511	130.02433	0.77
31	342.8875	140.01768	140.01522	2.46
32	307.3795	150.01777	150.01965	-1.88
33	277.5064	160.02581	160.02826	-2.45
34	252.1334	170.02917	170.03353	-4.36
35	230.3780	180.04850	180.03981	8.69
36	211.5883	190.04153	190.03996	1.57
37	195.2360	200.03538	200.03815	-2.77
38	180.9131	210.03431	210.03385	0.46
39	168.2739	220.03956	220.04322	-3.66
40	157.0908	230.03911	230.03889	0.22
41	147.1376	240.02863	240.02886	-0.23
42	138.2171	250.03609	250.03785	-1.76
43	130.2172	260.03361	260.02987	3.73
44	122.9923	270.04023	270.03609	4.14
45	116.4605	280.03188	280.03122	0.66
46	110.5161	290.04536	290.04886	-3.50
47	105.1110	300.04254	300.04965	-7.10
48	100.1725	310.05283	310.05306	-0.23
49	97.86332	315.06032	315.05201	8.31
50	95.64867	320.05632	320.05857	-2.25
51	93.12094	326.05141	326.04712	4.29
52	91.49748	330.05447	330.05888	-4.41

Order of Fit = 8 RMS error of fit = 3.45 mK

Largest absolute error = 8.69 mK at data point no. 35

INTERPOLATION TABLE

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>	<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>
20.00	7449.53	-600.77	-1.6129	130.0	385.667	-4.6994	-1.5841
21.00	6886.91	-526.77	-1.6063	135.0	363.285	-4.2631	-1.5842
22.00	6391.88	-465.10	-1.6008	140.0	342.947	-3.8806	-1.5842
23.00	5953.43	-413.25	-1.5965	145.0	324.403	-3.5439	-1.5840
24.00	5562.79	-369.21	-1.5929	150.0	307.443	-3.2459	-1.5837
25.00	5212.87	-331.56	-1.5901	155.0	291.888	-2.9812	-1.5831
26.00	4897.94	-299.10	-1.5877	160.0	277.584	-2.7450	-1.5822
27.00	4613.24	-270.95	-1.5858	165.0	264.397	-2.5336	-1.5811
28.00	4354.84	-246.40	-1.5842	170.0	252.212	-2.3437	-1.5798
29.00	4119.45	-224.85	-1.5829	175.0	240.928	-2.1726	-1.5781
30.00	3904.29	-205.87	-1.5818	180.0	230.458	-2.0180	-1.5761
31.00	3707.00	-189.05	-1.5810	185.0	220.724	-1.8779	-1.5739
32.00	3525.56	-174.10	-1.5802	190.0	211.658	-1.7506	-1.5715
33.00	3358.27	-160.75	-1.5796	195.0	203.199	-1.6347	-1.5687
34.00	3203.60	-148.79	-1.5792	200.0	195.294	-1.5289	-1.5658
35.00	3060.28	-138.03	-1.5786	205.0	187.895	-1.4322	-1.5626
36.00	2927.18	-128.33	-1.5783	210.0	180.959	-1.3436	-1.5592
37.00	2803.31	-119.56	-1.5781	215.0	174.447	-1.2622	-1.5557
38.00	2687.79	-111.60	-1.5778	220.0	168.325	-1.1874	-1.5519
39.00	2579.87	-104.36	-1.5776	225.0	162.563	-1.1185	-1.5480
40.00	2478.87	-97.756	-1.5774	230.0	157.132	-1.0548	-1.5440
42.00	2295.26	-86.187	-1.5771	235.0	152.007	-0.99598	-1.5398
44.00	2132.90	-76.441	-1.5769	240.0	147.165	-0.94150	-1.5354
46.00	1988.52	-68.163	-1.5768	245.0	142.585	-0.89098	-1.5309
48.00	1859.46	-61.077	-1.5766	250.0	138.249	-0.84406	-1.5263
50.00	1743.55	-54.980	-1.5767	255.0	134.139	-0.80042	-1.5216
52.00	1639.00	-49.693	-1.5766	260.0	130.240	-0.75977	-1.5167
54.00	1544.33	-45.084	-1.5764	265.0	126.537	-0.72186	-1.5118
56.00	1458.28	-41.055	-1.5766	270.0	123.017	-0.68646	-1.5067
58.00	1379.79	-37.505	-1.5765	273.15	120.888	-0.66535	-1.5034
60.00	1307.98	-34.368	-1.5766	275.0	119.668	-0.65336	-1.5014
65.00	1152.91	-27.966	-1.5767	280.0	116.480	-0.62237	-1.4961
70.00	1025.76	-23.108	-1.5769	285.0	113.441	-0.59332	-1.4906
75.00	920.007	-19.348	-1.5773	290.0	110.544	-0.56606	-1.4850
77.35	876.306	-17.871	-1.5775	295.0	107.778	-0.54045	-1.4793
80.00	830.953	-16.388	-1.5777	300.0	105.137	-0.51637	-1.4734
85.00	755.144	-14.022	-1.5783	305.0	102.612	-0.49369	-1.4674
90.00	689.991	-12.104	-1.5789	310.0	100.198	-0.47232	-1.4613
95.00	633.525	-10.533	-1.5795	315.0	97.8868	-0.45217	-1.4551
100.0	584.205	-9.2347	-1.5807	320.0	95.6740	-0.43314	-1.4487
105.0	540.826	-8.1474	-1.5818	325.0	93.5537	-0.41515	-1.4422
110.0	502.450	-7.2287	-1.5826				
115.0	468.311	-6.4470	-1.5831				
120.0	437.794	-5.7772	-1.5835				
125.0	410.385	-5.1999	-1.5838				

THERMAL CYCLE TESTING

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Serial Number: X115143

Sensor Type: Cernox Resistor

This sensor was tested for repeatability through rapid thermal cycles from room temperature into liquid helium. During this test, the following four lead resistance values were recorded:

Approximately 295 K:	107 Ω
Liquid Nitrogen:	876 Ω
Liquid Helium:	172081 Ω

The nitrogen and helium values were recorded in OPEN dewars, so precision comparisons with calibration values or other thermal cycle test values should not be made.

Recommended Operating Parameters:

For sensors calibrated by Lake Shore, the current to the sensor is adjusted to maintain the sensor output voltage or power at the values listed on the Test Data page.

BREAKPOINTS CUBIC SPLINE FORMAT

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Sensor Model: CX-1080-CU-HT-20L
Serial Number: X115143
Data Format: 7 (Ohms/Kelvin)
Setpoint Limit: 325

Measurement (ohms)	Temp (K)	Curvature
9.14975E+01	3.30059E+02	5.22599E-02
9.31209E+01	3.26047E+02	4.96202E-02
9.56487E+01	3.20059E+02	4.55103E-02
9.78633E+01	3.15052E+02	4.23859E-02
1.00173E+02	3.10053E+02	3.93343E-02
1.05111E+02	3.00050E+02	3.38638E-02
1.10516E+02	2.90049E+02	2.90644E-02
1.16461E+02	2.80031E+02	2.48201E-02
1.22992E+02	2.70036E+02	2.11018E-02
1.30217E+02	2.60030E+02	1.78408E-02
1.38217E+02	2.50038E+02	1.49987E-02
1.47138E+02	2.40029E+02	1.25237E-02
1.57091E+02	2.30039E+02	1.03862E-02
1.68274E+02	2.20043E+02	8.54465E-03
1.80913E+02	2.10034E+02	6.96719E-03
1.95236E+02	2.00038E+02	5.62903E-03
2.11588E+02	1.90040E+02	4.50094E-03
2.30378E+02	1.80040E+02	3.55868E-03
2.52133E+02	1.70034E+02	2.77918E-03
2.77506E+02	1.60028E+02	2.14172E-03
3.07380E+02	1.50020E+02	1.62605E-03
3.42887E+02	1.40015E+02	1.21409E-03
3.85552E+02	1.30024E+02	8.89045E-04
4.37588E+02	1.20036E+02	6.35741E-04
5.02141E+02	1.10043E+02	4.40708E-04
5.83854E+02	1.00038E+02	2.96354E-04
6.33123E+02	9.50376E+01	2.44261E-04
6.89490E+02	9.00414E+01	1.93774E-04
7.54463E+02	8.50486E+01	1.53298E-04
8.30162E+02	8.00483E+01	1.19103E-04
9.18925E+02	7.50560E+01	9.11528E-05
1.02433E+03	7.00621E+01	6.84335E-05
1.15125E+03	6.50595E+01	5.02566E-05
1.30555E+03	6.00707E+01	3.60360E-05
1.49703E+03	5.50760E+01	2.50342E-05
1.73929E+03	5.00776E+01	1.69245E-05
1.85428E+03	4.80850E+01	1.44532E-05
2.05300E+03	4.50785E+01	1.09798E-05
2.28868E+03	4.20766E+01	8.25273E-06
2.57112E+03	3.90841E+01	6.06306E-06
2.91718E+03	3.60782E+01	4.33692E-06
3.34688E+03	3.30711E+01	3.02650E-06
3.71622E+03	3.09513E+01	2.31290E-06
4.09085E+03	2.91279E+01	1.79631E-06
4.48030E+03	2.75026E+01	1.41535E-06
4.93115E+03	2.58896E+01	1.09902E-06
5.45638E+03	2.42928E+01	8.42021E-07
6.07765E+03	2.27045E+01	6.32990E-07
6.82923E+03	2.11103E+01	4.67015E-07
7.43041E+03	2.00319E+01	3.74728E-07
8.05577E+03	1.90552E+01	3.07034E-07
8.77434E+03	1.80796E+01	2.29250E-07



BREAKPOINTS 340 FORMAT

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Name: CX-1080-CU-HT-20L

Serial Number: X115143

Format: 4 ;Log Ohms/Kelvin

Limit: 325.0

Coefficient: 1 ;Negative

Point 1: 1.97103,325.000	Point 51: 2.79813, 95.500
Point 2: 1.98275,319.000	Point 52: 2.81633, 93.000
Point 3: 1.99372,313.500	Point 53: 2.83501, 90.500
Point 4: 2.00495,308.000	Point 54: 2.85422, 88.000
Point 5: 2.01643,302.500	Point 55: 2.87397, 85.500

Point 6: 2.02817,297.000	Point 56: 2.89431, 83.000
Point 7: 2.04019,291.500	Point 57: 2.91104, 81.000
Point 8: 2.05248,286.000	Point 58: 2.92817, 79.000
Point 9: 2.06507,280.500	Point 59: 2.94574, 77.000
Point 10: 2.07796,275.000	Point 60: 2.96376, 75.000

Point 11: 2.09115,269.500	Point 61: 2.98227, 73.000
Point 12: 2.10344,264.500	Point 62: 3.00130, 71.000
Point 13: 2.11599,259.500	Point 63: 3.02086, 69.000
Point 14: 2.12883,254.500	Point 64: 3.04100, 67.000
Point 15: 2.14197,249.500	Point 65: 3.06176, 65.000

Point 16: 2.15541,244.500	Point 66: 3.08315, 63.000
Point 17: 2.16917,239.500	Point 67: 3.10524, 61.000
Point 18: 2.18326,234.500	Point 68: 3.12805, 59.000
Point 19: 2.19770,229.500	Point 69: 3.14928, 57.200
Point 20: 2.21249,224.500	Point 70: 3.17116, 55.400

Point 21: 2.22765,219.500	Point 71: 3.19377, 53.600
Point 22: 2.24321,214.500	Point 72: 3.21716, 51.800
Point 23: 2.25916,209.500	Point 73: 3.24137, 50.000
Point 24: 2.27554,204.500	Point 74: 3.26647, 48.200
Point 25: 2.29067,200.000	Point 75: 3.28960, 46.600

Point 26: 2.30615,195.500	Point 76: 3.31352, 45.000
Point 27: 2.32202,191.000	Point 77: 3.33831, 43.400
Point 28: 2.33830,186.500	Point 78: 3.36403, 41.800
Point 29: 2.35500,182.000	Point 79: 3.39076, 40.200
Point 30: 2.37214,177.500	Point 80: 3.41681, 38.700

Point 31: 2.38973,173.000	Point 81: 3.44389, 37.200
Point 32: 2.40781,168.500	Point 82: 3.47209, 35.700
Point 33: 2.42639,164.000	Point 83: 3.49953, 34.300
Point 34: 2.44550,159.500	Point 84: 3.52810, 32.900
Point 35: 2.46517,155.000	Point 85: 3.55793, 31.500

Point 36: 2.48543,150.500	Point 86: 3.58689, 30.200
Point 37: 2.50397,146.500	Point 87: 3.61710, 28.900
Point 38: 2.52301,142.500	Point 88: 3.64876, 27.600
Point 39: 2.54260,138.500	Point 89: 3.67939, 26.400
Point 40: 2.56276,134.500	Point 90: 3.71145, 25.200

Point 41: 2.58353,130.500	Point 91: 3.74516, 24.000
Point 42: 2.60494,126.500	Point 92: 3.77767, 22.900
Point 43: 2.62704,122.500	Point 93: 3.81184, 21.800
Point 44: 2.64986,118.500	Point 94: 3.84791, 20.700
Point 45: 2.67347,114.500	Point 95: 3.87210, 20.000

Point 46: 2.69483,111.000
Point 47: 2.71684,107.500
Point 48: 2.73958,104.000
Point 49: 2.76308,100.500
Point 50: 2.78041, 98.000



BREAKPOINTS 91C/93C/330 FORMAT

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Interpolation Method: Lagrangian

Limit: 325.0 (Kelvin)

Format: 4 (Log Ohms/Kelvin)

Number of Breakpoints: 35

No.	Units	Temperature (K)	No.	Units	Temperature (K)
1	1.97106	325.0	21	3.18243	54.5
2	1.97299	324.0	22	3.24832	49.5
3	2.00291	309.0	23	3.28966	46.6
4	2.03471	294.0	24	3.34154	43.2
5	2.06858	279.0	25	3.38913	40.3
6	2.10470	264.0	26	3.43847	37.5
7	2.14332	249.0	27	3.48772	34.9
8	2.18472	234.0	28	3.53659	32.5
9	2.22922	219.0	29	3.58471	30.3
10	2.27723	204.0	30	3.63408	28.2
11	2.32924	189.0	31	3.68210	26.3
12	2.38582	174.0	32	3.73103	24.5
13	2.44770	159.0	33	3.77779	22.9
14	2.51585	144.0	34	3.82487	21.4
15	2.59152	129.0	35	3.87213	20.0
16	2.67654	114.0			
17	2.77346	99.0			
18	2.88614	84.0			
19	3.02090	69.0			
20	3.12233	59.5			

Temperature for Resistance Decades:

Res. (Ohms)	Temp. (K)
100	310.419
1000	71.133

BREAKPOINTS 234 FORMAT

Calibration Report: 935212

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115143

Temperature Range: 20.0 K to 325 K

Maximum Temperature Error:

1.4 - 10 K: -
10 - 20 K: 0.006 K
20 - 40 K: 0.010 K
40 - 100 K: 0.023 K
> 100 K: 0.030 K

BP #	Temp. (K)	Res. (W)	Log10 Res.	BP #	Temp. (K)	Res. (W)	Log10 Res.
1	320.404	95.49926	1.980	41	98.062	602.5596	2.780
2	310.419	100.0000	2.000	42	95.245	630.9573	2.800
3	300.824	104.7129	2.020	43	92.507	660.6934	2.820
4	291.595	109.6478	2.040	44	89.848	691.8310	2.840
5	282.710	114.8154	2.060	45	87.265	724.4360	2.860
6	274.149	120.2264	2.080	46	84.756	758.5776	2.880
7	265.897	125.8925	2.100	47	82.318	794.3282	2.900
8	257.935	131.8257	2.120	48	79.951	831.7638	2.920
9	250.250	138.0384	2.140	49	77.650	870.9636	2.940
10	242.828	144.5440	2.160	50	75.416	912.0108	2.960
11	235.656	151.3561	2.180	51	73.246	954.9926	2.980
12	228.723	158.4893	2.200	52	71.138	1000.000	3.000
13	222.017	165.9587	2.220	53	67.102	1096.478	3.040
14	215.530	173.7801	2.240	54	63.295	1202.264	3.080
15	209.251	181.9701	2.260	55	59.703	1318.257	3.120
16	203.171	190.5461	2.280	56	56.315	1445.440	3.160
17	197.282	199.5262	2.300	57	53.119	1584.893	3.200
18	191.576	208.9296	2.320	58	50.105	1737.801	3.240
19	186.045	218.7762	2.340	59	47.262	1905.461	3.280
20	180.683	229.0868	2.360	60	44.580	2089.296	3.320
21	175.483	239.8833	2.380	61	42.051	2290.868	3.360
22	170.438	251.1886	2.400	62	39.666	2511.886	3.400
23	165.543	263.0268	2.420	63	37.416	2754.229	3.440
24	160.792	275.4229	2.440	64	35.295	3019.952	3.480
25	156.181	288.4032	2.460	65	33.295	3311.311	3.520
26	151.703	301.9952	2.480	66	31.410	3630.781	3.560
27	147.355	316.2278	2.500	67	29.633	3981.072	3.600
28	143.133	331.1311	2.520	68	27.958	4365.158	3.640
29	139.032	346.7369	2.540	69	26.380	4786.301	3.680
30	135.049	363.0781	2.560	70	24.894	5248.075	3.720
31	131.179	380.1894	2.580	71	23.495	5754.399	3.760
32	127.420	398.1072	2.600	72	22.179	6309.573	3.800
33	123.769	416.8694	2.620	73	20.941	6918.310	3.840
34	120.222	436.5158	2.640	74	19.777	7585.776	3.880
35	116.776	457.0882	2.660	75	18.683	8317.638	3.920
36	113.428	478.6301	2.680				
37	110.175	501.1872	2.700				
38	107.015	524.8075	2.720				
39	103.944	549.5409	2.740				
40	100.961	575.4399	2.760				